Servlet and JSP

Problem Statement:

You are asked to develop a Java Web Application called Employee Management System using JSP, Servlets, Hibernate with JPA, a database (PostgreSQL), and Tomcat.

The application must manage employee records in the database and demonstrate the usage of Servlet, JSP, ServletContext, ServletConfig, HttpSession, Cookies, RequestDispatcher, sendRedirect, Welcome-file, and Load-on-startup.

Application Requirements

1. Home Page (index.jsp)

- Define it as the welcome file in web.xml.
- Provide navigation links to:
  - View All Employees (employee)
  - Add New Employee (employee?action=create)

2. Add New Employee

- Create a JSP form (employeeForm.jsp) to input:
  - Name
  - Email
  - Department (pre-fill with a default department from ServletConfig if empty).
- On form submission (POST):
  - A Servlet should:
    - Create an Employee entity object (the unique ID is auto-generated by the database).
    - Save it to the database using Hibernate with JPA (EntityManager).
  - Use sendRedirect("employee") to return to the list view.

3. View All Employees (employee)

- Display all employee records (fetched from the database using JPA) in a table.
- For each record, provide:
  - Edit link → employee?action=edit&id=EMP_ID
  - Delete link → employee?action=delete&id=EMP_ID (with confirmation).
- At the bottom of the page:
  - Display the last updated employee ID (from HttpSession).
  - Display the last updated employee name (from a Cookie).

4. Edit Employee

- When user clicks Edit:
  - The servlet must load the employee details by ID from the database using JPA.
  - Use RequestDispatcher to forward them to the same employeeForm.jsp form.
  - Pre-fill the form fields with existing data.
- On submission:
  - Update the employee record in the database using JPA.
  - Store:

        - Employee ID in HttpSession.
        - Employee name in a Cookie.
    - Redirect back to the employee list.

5. Delete Employee

- When user clicks Delete:
    - Remove the employee from the database using JPA.
    - Use sendRedirect("employee") to refresh the list.